Check 1 (Mean): why $\widehat{\mathrm{se}}_{\mathrm{jack}} = s/\sqrt{n}$

Let $\hat{\theta} = \bar{x}$ and $\hat{\theta}_{(i)} = \bar{x}_{(i)}$.

$$\bar{x}_{(i)} = \frac{1}{n-1}\sum_{j\neq i} x_j = \frac{n\bar{x} - x_i}{n-1}.$$

Average the leave-one-out means:

$$\hat{\theta}_{(\cdot)} = \frac{1}{n}\sum_{i=1}^{n} \bar{x}_{(i)} = \bar{x}_{\cdot}$$

Hence

$$\bar{x}_{(i)} - \bar{x} = \frac{\bar{x} - x_i}{n-1} \quad \Rightarrow \quad \sum_{i=1}^{n}(\bar{x}_{(i)} - \bar{x})^2 = \frac{1}{(n-1)^2}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Plugging into the jackknife SE formula gives

$$\widehat{\mathrm{se}}^2_{\mathrm{jack}} = \frac{n-1}{n}\cdot\frac{1}{(n-1)^2}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n(n-1)}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{s^2}{n}.$$

Where the Jackknife Fails – and Bootstrap vs Jackknife

Failure for non-smooth statistics

For the **sample median**, deleting one point moves $\hat{\theta}_{(i)}$ to one of only two values, so $\widehat{\text{se}}_{\text{jack}}$ is *inconsistent* – it does not converge to the true standard error. (The delete-d jackknife with $d \to \infty$ repairs this.)

$\hat{\theta}$ = sample median $\Rightarrow$

Copy 1 $\to$ $X_2, X_3 \ldots X_n \to$

2 $\to$ $X_1, X_3 \ldots X_n \to$

$\vdots$

$n \to$ $X_1 X_2 \ldots X_{n-1} \to$

a, b, a, b, ... a, b

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	Jackknife	Bootstrap
Samples	n deterministic leave-one-out	B random resamples
Randomness in answer	none (reproducible)	Monte Carlo error
Smooth statistics	good (a linear approximation)	good
Median, quantiles	fails	works
Confidence intervals	not directly	yes

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- Today the jackknife's main roles: cheap bias correction.

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1. Motivation: Inference When Formulas Run Out
2. The Bootstrap Principle
3. Parametric vs Nonparametric Bootstrap
4. Bootstrap Confidence Intervals
5. The Jackknife (Brief)
6. Monte Carlo Hypothesis Testing and Permutation Tests
   - Monte Carlo p-Values
   - Parametric Bootstrap Tests and Simulated Power
   - Permutation Tests
   - Example: A Two-Sample Permutation Test
   - Exchangeability: Permutation vs Bootstrap Tests
7. Unit Summary

Testing Without a Null Distribution Table

- Classical testing needs the **null distribution** of the statistic T: t, χ^2, F tables.

F, $\mu \rightarrow$ pop^n mean parameter, σ^2: given

$H_0: \mu = 10$, $H_1: \mu \neq 10$

H_0: null hypothesis; H_1: Altern.

$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$X_1 \cdots X_n \sim F$

$\hookrightarrow$ test statistic $\Rightarrow Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$

$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \Rightarrow$ under $H_0 \Rightarrow Z = \frac{\bar{X} - 10}{\sigma/\sqrt{n}}$

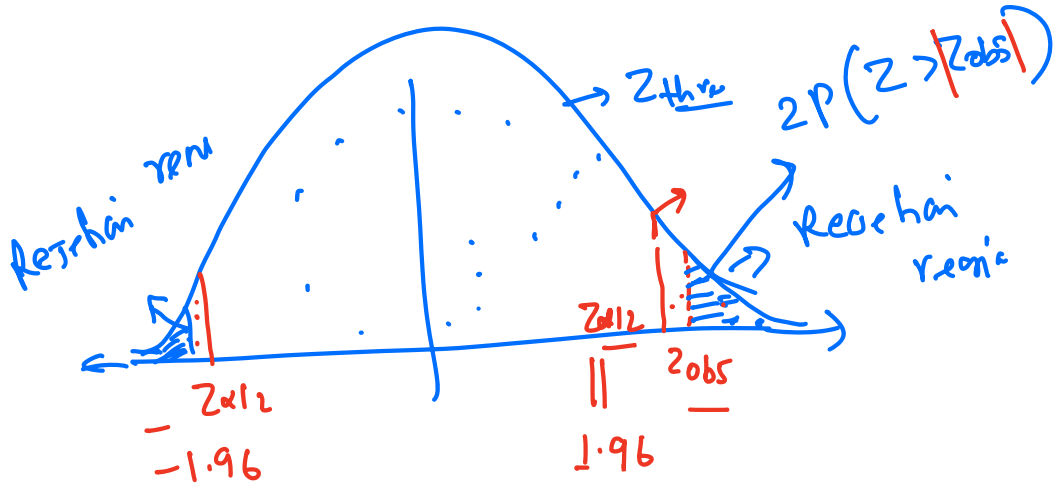
$Z_{obs} = \frac{\frac{1}{n}\sum_{i=1}^{n} x_i - 10}{\sigma/\sqrt{n}}$

$Z_{obs} = (\bar{X} - 10)/\sigma/\sqrt{n}$ under H_0 ✓

$\Rightarrow \alpha = 5\% = 0.05$

$Z_{theo} \sim N(0,1)$

Z_α



$\Rightarrow Z < -|Z_{obs}|$ or $Z > |Z_{obs}|$ → Reject

$\Rightarrow Z_{obs} = 0.5$ → $Z_{obs} = 2.5$ → Reject

-2.8 → Reject ($\alpha = 10\%$)

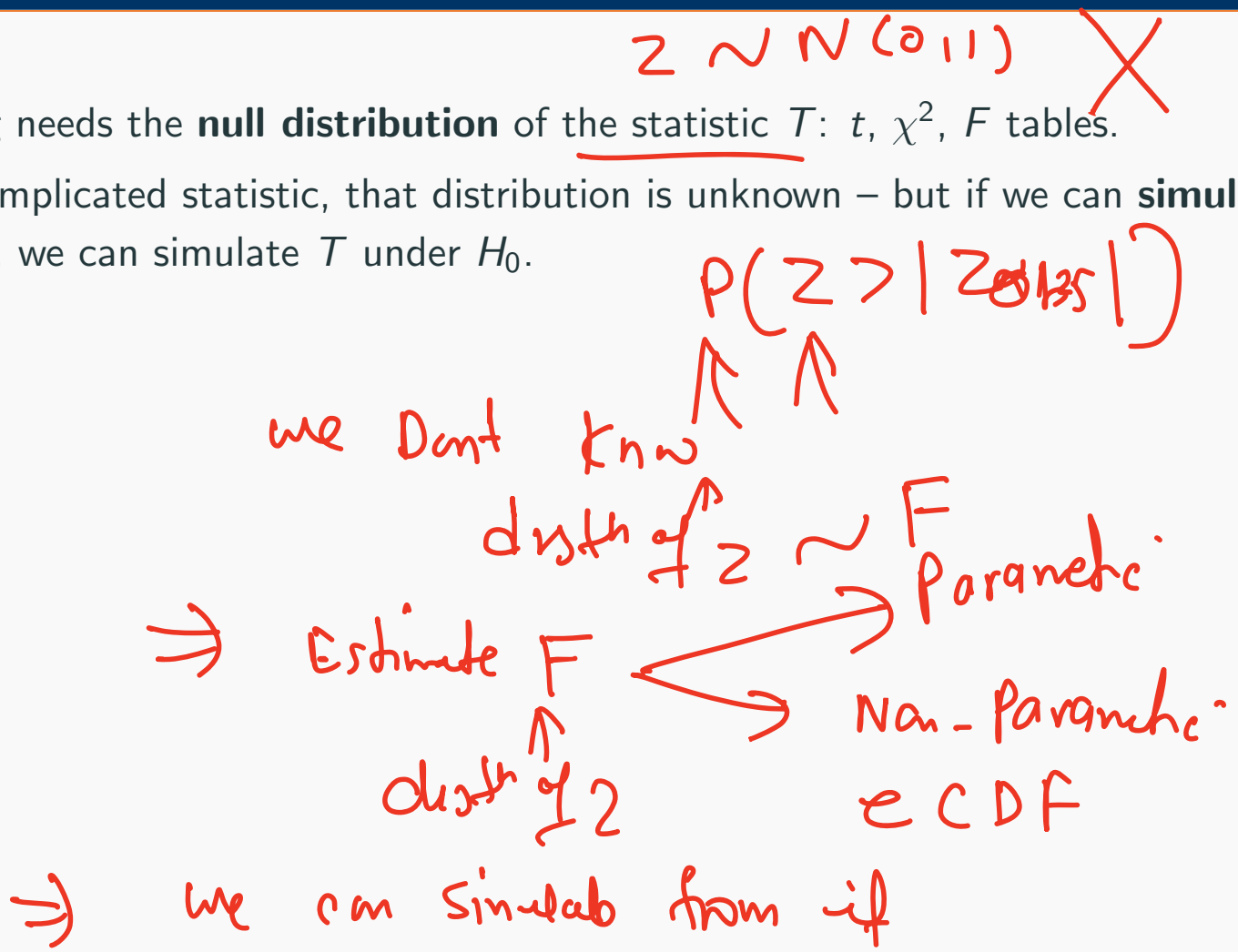
$2P(Z > |Z_{obs}|) = P = 0.02$

$P < \alpha$ → Reject

$P > \alpha$ → failed to reject the H_0

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Algorithm: Monte Carlo test

1. Compute $t_{\text{obs}} = T(\boldsymbol{x})$.
2. For $b = 1, \ldots, B$, simulate $\boldsymbol{x}^{*b}$ under H_0; compute T_b^*.
3. Return

$$\hat{p} = \frac{1 + \#\{b : T_b^* \geq t_{\text{obs}}\}}{B + 1}.$$

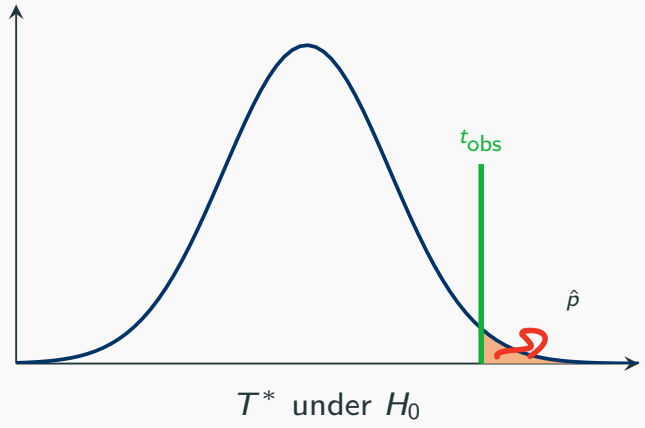
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Simulating Size and Power

- Monte Carlo also tells us how a test **behaves**, before we ever see real data.

α: size of a test

$\alpha = 0.05 \rightarrow$ an you can get an Idea about it based on the bootstrap

Simulating Size and Power

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Algorithm: Simulated rejection rate

For a data-generating distribution G:

1. Repeat N times: simulate a dataset from G, run the test, record whether it rejects at level α.
2. Rejection rate $\hat{\pi}(G) = (\#\text{rejections})/N$, with se $\approx \sqrt{\hat{\pi}(1-\hat{\pi})/N}$.

If G satisfies H_0, $\hat{\pi}$ estimates the **size**; otherwise it estimates the **power**.

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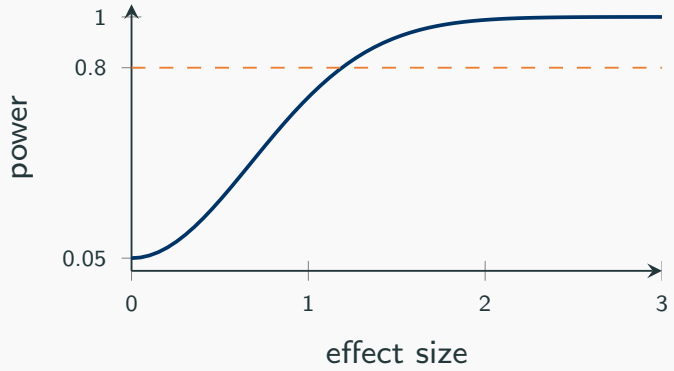
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- Use parallel-safe, **seeded** streams (Module 1) so results are reproducible.
- Spend the budget where error matters: increasing B beyond the point where Monte Carlo error is small compared with the statistical error buys nothing.

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Putting It All Together

Method	Core idea	Best suited for
Nonparametric bootstrap	Resample data with replacement	SE, bias, CI of any statistic
Parametric bootstrap	Simulate from fitted model	Small n, tails, MLE-based inference
Jackknife	Leave one out	Bias correction, BCa acceleration
Monte Carlo test	Simulate T under H_0	Non-standard statistics, composite H_0